Mathematical Analysis

Functions, convergence, calculus and optimization for Data Science.

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Why analysis for Data Science?

An algorithm produces a sequence.

A model transforms inputs into predictions.

A fitted parameter minimizes a loss.

> Do the estimates converge? How sensitive is the output? Does an optimum exist?

The route through this course

Numbers and functions

Sequences, limits and continuity

Derivatives and local approximations

Integrals and accumulation

Optimization in one and several variables

Course Outline

Numbers and algebraic foundations

Functions and bounds

Sequences and iterative algorithms

Limits, continuity and existence

Derivatives and local approximation

Integration and accumulation

Optimization in one variable

Several variables and optimization

Course Outline

Numbers and algebraic foundations

Functions and bounds

Sequences and iterative algorithms

Limits, continuity and existence

Derivatives and local approximation

Integration and accumulation

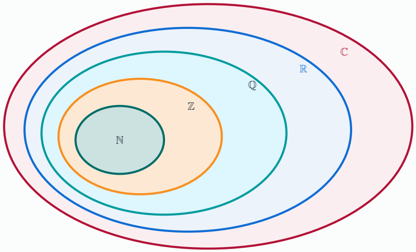
Optimization in one variable

Several variables and optimization

Number systems enlarge what we can express

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}.$$

Set	Examples
$\mathbb{N}$	$0, 1, 2, \ldots$
$\mathbb{Z} \setminus \mathbb{N}$	$-1, -2$
$\mathbb{Q} \setminus \mathbb{Z}$	$1/2, -7/3$
$\mathbb{R} \setminus \mathbb{Q}$	$\sqrt{2}, \pi$
$\mathbb{C} \setminus \mathbb{R}$	$\mathrm{i}, 2 + 3\mathrm{i}$



Completeness makes limits possible in the real line

DEFINITION — The least-upper-bound property

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

EXAMPLE — A gap in the rational numbers

The rational set $A = \{q \in \mathbb{Q} : q \geq 0,\ q^2 < 2\}$ is nonempty and bounded. Its supremum in $\mathbb{R}$ is $\sqrt{2}$, which does not belong to $\mathbb{Q}$.

Completeness ensures that bounded monotone approximations have a real limit.

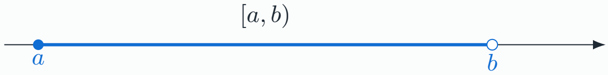
Density and completeness are different: rational numbers approximate every real number, but rational limits need not be rational.

Intervals specify which endpoints belong to the domain

DEFINITION — Intervals, for $a < b$

$$(a, b) = \{x : a < x < b\}, \qquad [a, b] = \{x : a \le x \le b\}.$$

The intervals $[a, b)$ and $(a, b]$ include exactly one endpoint.



Unbounded intervals include $(a, +\infty)$ and $(-\infty, b]$.

Check the assumptions

Infinity is not a real endpoint: brackets are never closed at $\pm\infty$. Endpoint inclusion matters for extrema.

Groups, rings and fields organize valid operations

Structure	Defining operations	Example
Group	Associative operation, identity and inverses	$(\mathbb{Z}, +)$
Ring	Abelian additive group; associative multiplication with identity; distributivity	$(\mathbb{Z}, +, \times)$
Field	Commutative ring with $0 \neq 1$; every nonzero element has a multiplicative inverse	$\mathbb{Q}, \mathbb{R}, \mathbb{C}$

EXAMPLE — Division changes the required structure

$2x = 1$ has no integer solution, but has the rational solution $x = 1/2$.

Complex numbers close polynomial equations

DEFINITION — Complex numbers

$\mathbb{C} = \{x + \mathrm{i}y : x, y \in \mathbb{R}\}$, where $\mathrm{i}^2 = -1$.

THEOREM — Fundamental theorem of algebra

Every nonconstant polynomial in $\mathbb{C}[X]$ has a complex root. Consequently, a degree-n polynomial factors as

$$P(X) = a_n \prod_{j=1}^{n} (X - \alpha_j), \qquad \alpha_j \in \mathbb{C},$$

with roots counted according to multiplicity.

EXAMPLE — A real polynomial with nonreal roots

$X^2 + 1 = (X - \mathrm{i})(X + \mathrm{i})$.

Course Outline

A function includes its domain and codomain

DEFINITION — Function

$$f : E \to F, \qquad x \mapsto f(x).$$

Every $x \in E$ has exactly one image in F.

EXAMPLE — The same formula on different domains

$f(x) = x^2$ is defined on $\mathbb{R}$, but its restriction to $[0, +\infty)$ has different invertibility properties.

$$\mathrm{Im}(f) = f(E) = \{f(x) : x \in E\} \subseteq F.$$

The image is what the function actually reaches; the codomain is where its values are required to lie.

A formula must be paired with a valid domain

Formula	Real domain	Image
x^2	$\mathbb{R}$	$[0, +\infty)$
$\sqrt{x}$	$[0, +\infty)$	$[0, +\infty)$
$\log x$	$(0, +\infty)$	$\mathbb{R}$
$1/(x-1)$	$\mathbb{R} \setminus \{1\}$	$\mathbb{R} \setminus \{0\}$

EXAMPLE — Two simultaneous restrictions

For $f(x) = \sqrt{3-x}/(x-1)$, the domain is

$$D = (-\infty, 3] \setminus \{1\}.$$

The square root requires $x \leq 3$; the denominator requires $x \neq 1$.

The supremum need not be an attained maximum

DEFINITION — Upper bound and supremum

M is an upper bound of A if $x \le M$ for every $x \in A$. For nonempty bounded-above A, $\sup A$ is its least upper bound.

$$M = \sup A \iff \begin{cases} x \le M & \text{for every } x \in A, \\ \forall \varepsilon > 0,\ \exists x \in A :\ x > M - \varepsilon. \end{cases}$$

EXAMPLE — Open and closed intervals

$\sup(0,1) = 1$, but $(0,1)$ has no maximum.
$\max[0,1] = \sup[0,1] = 1$ because 1 belongs to the set.

The infimum is the greatest lower bound; a minimum is an attained infimum.

Bounds of a function are bounds of its image

$$\sup_{x \in D} f(x) = \sup f(D).$$

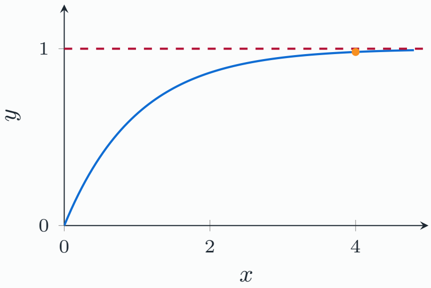
For $f(x) = 1 - e^{-x}$:

$$\max_{[0,4]} f = 1 - e^{-4} < 1.$$

On the unbounded domain $[0, +\infty)$,

$$\sup f = 1, \qquad f(x) < 1 \text{ for all } x.$$

The domain determines whether the bound is attained.



Monotonicity compares outputs in input order

DEFINITION — Increasing and decreasing functions

On an interval I, f is increasing if

$$x < y \implies f(x) \leq f(y).$$

It is strictly increasing if the output inequality is strict. Reverse the output inequality to define decreasing functions.

EXAMPLE — Monotonicity depends on the interval

x^2 decreases on $(-\infty, 0]$ and increases on $[0, +\infty)$. It is not monotone on all of $\mathbb{R}$.

Strict monotonicity rules out repeated outputs and therefore gives injectivity.

Operations and composition build new functions

For functions on a common domain,

$$(af + bg)(x) = af(x) + bg(x), \qquad (fg)(x) = f(x)g(x).$$

The quotient f/g is defined where $g(x) \neq 0$.

DEFINITION — Composition

If $g : E \to F$ and $f : F \to G$, then

$$(f \circ g)(x) = f(g(x)).$$

EXAMPLE — Order matters

If $f(x) = x^2$ and $g(x) = x + 1$, then

$$(f \circ g)(x) = (x + 1)^2, \qquad (g \circ f)(x) = x^2 + 1.$$

Inverse images collect all inputs reaching a set

DEFINITION — Inverse image

For $f : E \to F$ and $B \subseteq F$,

$$f^{-1}(B) = \{x \in E : f(x) \in B\}.$$

EXAMPLE — Squaring on the real line

For $f(x) = x^2$,

$$f^{-1}(\{4\}) = \{-2, 2\}, \qquad f^{-1}([1, 4]) = [-2, -1] \cup [1, 2].$$

Also $f^{-1}(\{-1\}) = \varnothing$.

Check the assumptions

The notation $f^{-1}(B)$ does not require an inverse function. An inverse image is a set and can contain several points or none.

Bijections can be inverted as functions

DEFINITION — Injective, surjective, bijective

Injective: $f(x) = f(y) \implies x = y$.
Surjective onto F: every $y \in F$ has a preimage.
Bijective: both properties hold; every output has exactly one preimage.

EXAMPLE — Choosing the right domain and codomain

$x \mapsto x^2$ is not injective on $\mathbb{R}$ and not surjective onto $\mathbb{R}$. But $f : [0, +\infty) \to [0, +\infty)$, $f(x) = x^2$, is bijective with inverse $\sqrt{\cdot}$.

$$f^{-1} \circ f = \mathrm{Id}_E, \qquad f \circ f^{-1} = \mathrm{Id}_F\,.$$

Course Outline

Numbers and algebraic foundations

Functions and bounds

Sequences and iterative algorithms

Limits, continuity and existence

Derivatives and local approximation

Integration and accumulation

Optimization in one variable

Several variables and optimization

A sequence is a function on the natural numbers

DEFINITION — Real sequence

$$u : \mathbb{N} \to \mathbb{R}, \qquad n \mapsto u_n.$$

The index describes an iteration, observation or time step.

Increasing means $u_{n+1} \geq u_n$; decreasing means $u_{n+1} \leq u_n$.

DEFINITION — Bounded sequence

There exist $m, M \in \mathbb{R}$ such that $m \leq u_n \leq M$ for every n.

EXAMPLE — Bounded does not mean convergent

$u_n = (-1)^n$ is bounded between -1 and 1, but keeps oscillating.

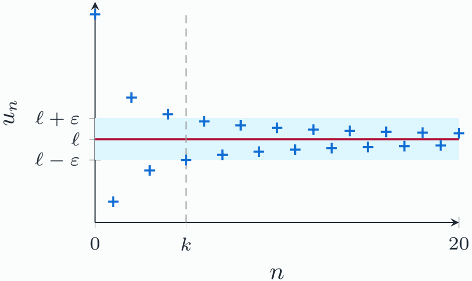
Convergence means eventually staying close

DEFINITION — Finite limit

$u_n \to \ell$ if, for every $\varepsilon > 0$, there exists k such that

$$n \geq k \implies |u_n - \ell| \leq \varepsilon.$$

The index k may depend on ε. The sequence must **stay** in the band, not merely enter it occasionally.



$u_n = 2 + 3(-1)^n/(n+1)$; $\varepsilon = 0.5$, $k = 5$.

An infinite limit crosses every fixed threshold

DEFINITION — Divergence to infinity

$$u_n \to +\infty \iff \forall M \in \mathbb{R},\ \exists k,\ \forall n \geq k:\ u_n \geq M.$$

For $u_n \to -\infty$, replace $u_n \geq M$ by $u_n \leq M$.

EXAMPLE — Three classical limits

$$\frac{1}{n+1} \to 0, \qquad \frac{2n+1}{n+1} = 2 - \frac{1}{n+1} \to 2, \qquad \sqrt{n+1} \to +\infty.$$

For the last sequence, choose $k+1 \geq \max(M,0)^2$.

Check the assumptions

An unbounded sequence need not tend to infinity: $(-1)^n n$ has neither a finite nor an infinite limit.

Limit rules need determinate expressions

If $u_n \to u$ and $v_n \to v$ are finite limits,

$$u_n + v_n \to u + v, \quad u_n v_n \to uv, \quad \frac{u_n}{v_n} \to \frac{u}{v} \quad (v \neq 0).$$

Check the assumptions

The seven standard indeterminate forms

$$\frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad \infty - \infty, \qquad 0 \cdot \infty, \qquad 1^{\infty}, \qquad 0^{0}, \qquad \infty^{0}.$$

These describe limiting behaviour, not arithmetic with infinity.

$$\frac{n}{n^2} \to 0, \qquad \frac{n}{n} \to 1, \qquad \frac{n^2}{n} \to +\infty.$$

All three have the form ∞/∞. Relative growth decides the answer.

Recursive sequences repeatedly apply the same map

DEFINITION — First-order recurrence

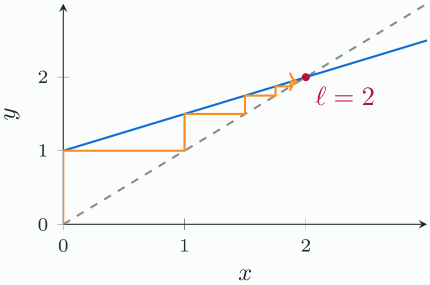
Let $f : D \to D$ and $u_0 \in D$.

$$u_{n+1} = f(u_n), \qquad n \geq 0.$$

The invariant domain D makes every step well-defined.

Vertical step: evaluate f.
Horizontal step: transfer to $y = x$.

$$f(x) = 1 + x/2, \quad u_0 = 0.$$



Induction propagates a property to every iteration

PROPOSITION — Mathematical induction

If $P(0)$ holds and $P(n) \implies P(n+1)$ for every $n \geq 0$, then $P(n)$ holds for all $n \in \mathbb{N}$.

EXAMPLE — An invariant interval

For $u_{n+1} = 1 + u_n/2$ and $u_0 = 0$, prove $0 \leq u_n \leq 2$.

Proof idea. The claim holds at $n = 0$. If $0 \leq u_n \leq 2$, then

$$1 \leq 1 + u_n/2 \leq 2.$$

Hence it holds at the next step and therefore at every step.

An invariant bound is often the first step in a convergence proof.

Monotone bounded sequences converge

THEOREM — Monotone convergence

An increasing sequence bounded above converges to $\sup_n u_n$. A decreasing sequence bounded below converges to $\inf_n u_n$.

Proof idea. In the increasing case, let $\ell = \sup_n u_n$. For $\varepsilon > 0$, some u_k satisfies

$$\ell - \varepsilon < u_k \leq \ell.$$

For every $n \geq k$, monotonicity gives

$$\ell - \varepsilon < u_k \leq u_n \leq \ell,$$

so $|u_n - \ell| < \varepsilon$.

Completeness supplies a candidate limit; monotonicity forces all later terms towards it.

Geometric decay quantifies the convergence rate

PROPOSITION — Geometric sequences

For $|q| < 1$, $cq^n \to 0$, and for $q \neq 1$,

$$\sum_{j=0}^{n} cq^j = c\frac{1 - q^{n+1}}{1 - q}.$$

EXAMPLE — Exact error in the affine iteration

For $u_{n+1} = 1 + u_n/2$ with $u_0 = 0$,

$$u_{n+1} - 2 = \tfrac{1}{2}(u_n - 2) \quad \Longrightarrow \quad u_n = 2 - 2^{1-n}.$$

The error is halved at each step.

For $q > 1$, magnitudes grow; for $q \leq -1$, nonzero geometric sequences oscillate without a limit.

Course Outline

Interior, closure and boundary

DEFINITION — Points relative to $A \subseteq \mathbb{R}^d$

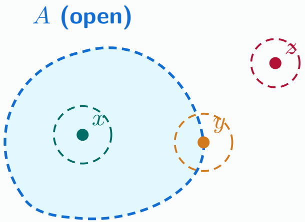
For $x \in \mathbb{R}^d$, let $B(x,r)$ denote the open ball of centre x and radius $r > 0$.

Interior: $x \in A^\circ$ if some ball $B(x,r)$ lies entirely in A.

Closure: $x \in \overline{A}$ if every ball $B(x,r)$ meets A.

Boundary: $x \in \partial A$ if every ball $B(x,r)$ meets both A and $\mathbb{R}^d \setminus A$.

$$\partial A = \overline{A} \setminus A^\circ.$$



$x \in A^\circ$: an interior point

$y \in \partial A$: a boundary point

$z \notin \overline{A}$: an exterior point

Both x and y belong to $\overline{A}$, but here $y \notin A$.

EXAMPLE — A half-open interval

For $A = [0,1)$: $A^\circ = (0,1)$, $\quad \overline{A} = [0,1]$, $\quad \partial A = \{0,1\}$.

At an interior point, sufficiently small perturbations are feasible in every direction; at a boundary point, this

A function limit describes nearby values

DEFINITION — Limit at an accumulation point

Let a be an accumulation point of D. Then $f(x) \to \ell$ as $x \to a$ if

$$\forall \varepsilon > 0,\ \exists \eta > 0,\ \forall x \in D : \quad 0 < |x - a| < \eta \implies |f(x) - \ell| < \varepsilon.$$

The puncture $0 < |x - a|$ excludes the point itself.

EXAMPLE — A removable gap

$$f(x) = \frac{x^2 - 1}{x - 1} = x + 1 \quad (x \neq 1), \qquad \lim_{x \to 1} f(x) = 2.$$

The limit exists although the original formula is undefined at 1.

Sequential limits provide a useful test

THEOREM — Sequential characterization

$f(x) \to \ell$ as $x \to a$ if and only if $f(x_n) \to \ell$ for every sequence $x_n \in D \setminus \{a\}$ with $x_n \to a$.

EXAMPLE — Disproving a limit

For $f(x) = \sin(1/x)$, choose

$$x_n = \frac{1}{\pi/2 + 2\pi n}, \qquad y_n = \frac{1}{3\pi/2 + 2\pi n}.$$

Both approach zero, but $f(x_n) = 1$ and $f(y_n) = -1$.

One suitable sequence can disprove a proposed limit. Proving a limit requires control of every approach.

Limits are unique and obey conditional rules

Proof idea. If $L \neq M$ were two limits, choose $\varepsilon = |L - M|/3$. At a sufficiently close domain point,

$$|L - M| \leq |L - f(x)| + |f(x) - M| < 2\varepsilon < |L - M|,$$

a contradiction.

PROPOSITION — Algebra and composition

Finite limits commute with sums and products; quotients require a nonzero denominator limit. If $g(x) \to \ell$ and f is continuous at ℓ, then $f(g(x)) \to f(\ell)$.

Do not substitute into a discontinuous outer function without checking additional conditions.

Continuity matches the limit to the actual value

DEFINITION — Continuity at $a \in D$

f is continuous at a if $f(x) \to f(a)$ as $x \to a$ within D. At an interior point of an interval, both one-sided limits must equal $f(a)$.

EXAMPLE — A jump at zero

$$f(x) = \begin{cases} x^2, & x < 0, \\ 1, & x \geq 0. \end{cases} \qquad \lim_{x \to 0^-} f(x) = 0, \quad \lim_{x \to 0^+} f(x) = 1.$$

Sums, products and compositions of continuous functions remain continuous on their domains; quotients also require a nonzero denominator.

Monotone functions have one-sided limiting behaviour

THEOREM — Monotone limit at infinity

If $f : [a, +\infty) \to \mathbb{R}$ is increasing and bounded above, then

$$\lim_{x \to +\infty} f(x) = \sup_{x \geq a} f(x).$$

If it is unbounded above, the limit is $+\infty$.

Proof idea. Let $L = \sup f$. For every $\varepsilon > 0$, choose X with $f(X) > L - \varepsilon$. Then $x \geq X$ gives $L - \varepsilon < f(X) \leq f(x) \leq L$.

EXAMPLE — Saturation

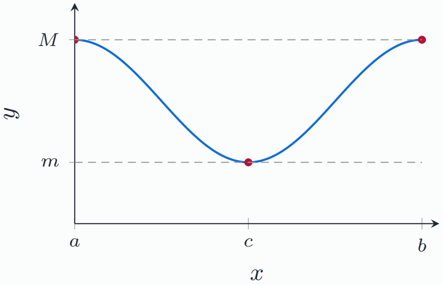
$f(x) = 1 - e^{-x}$ on $[0, +\infty)$ increases towards 1 without attaining it.

Compactness and continuity ensure attained extrema

THEOREM — Weierstrass

A continuous function on $[a, b]$, $a < b$, is bounded and attains its minimum and maximum.

Proof idea. The continuous image of the compact interval is compact, hence closed and bounded. It contains its supremum and infimum.



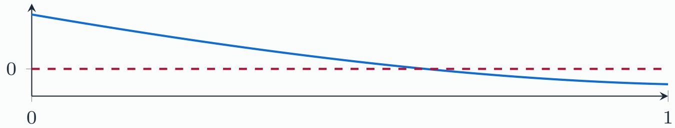
Continuity guarantees intermediate values

THEOREM — Intermediate value theorem

For continuous $f : [a, b] \to \mathbb{R}$, every value between $f(a)$ and $f(b)$ is attained at some $c \in [a, b]$.

EXAMPLE — Existence of a root

Let $g(x) = e^x - 3x$. Then $g(0) = 1 > 0$ and $g(1) = e - 3 < 0$. So there is a root in $(0, 1)$.



Strict monotonicity turns existence into uniqueness

THEOREM — Bijection theorem

If $f : I \to \mathbb{R}$ is continuous and strictly monotone on an interval, then $f : I \to f(I)$ is bijective. Its inverse is continuous and has the same monotonicity.

EXAMPLE — Revisiting $e^x = 3x$ on $(0, 1)$

$g'(x) = e^x - 3 < 0$ on $[0, 1]$, so the root established by the intermediate value theorem is unique there.

Existence and uniqueness are separate conclusions with separate justifications.

A continuous self-map of an interval has a fixed point

DEFINITION — Fixed point

ℓ is a fixed point of f when $f(\ell) = \ell$.

THEOREM — Interval fixed-point theorem

Every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Proof idea. Set $g(x) = f(x) - x$. Then $g(a) \geq 0$ and $g(b) \leq 0$. The intermediate value theorem gives $g(\ell) = 0$.

Check the assumptions

Existence does not ensure convergence of iteration. For $f(x) = 1 - x$ on $[0, 1]$, the fixed point is $1/2$, but iteration from 0 alternates between 0 and 1.

A contraction gives uniqueness and geometric convergence

THEOREM — Banach's theorem on a closed interval

Let $f : [a,b] \to [a,b]$ satisfy $|f(x) - f(y)| \leq q|x-y|$ with $0 \leq q < 1$. There is a unique fixed point ℓ, and every iteration satisfies

$$|u_n - \ell| \leq q^n |u_0 - \ell| \longrightarrow 0.$$

Proof idea. Lipschitz continuity and the interval theorem give existence. Two fixed points satisfy $|\ell_1 - \ell_2| \leq q|\ell_1 - \ell_2|$, hence coincide. Apply the contraction inequality repeatedly for the error bound.

This theorem provides all three: existence, uniqueness and convergence of an algorithm.

Course Outline

A derivative measures instantaneous change

DEFINITION — Derivative at an interior point

$$f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h},$$

provided this limit exists and is finite.

EXAMPLE — Position and velocity

If $s(t)$ is position, $[s(t+h) - s(t)]/h$ is average velocity. Its limit $s'(t)$ is instantaneous velocity.

Differentiability is equivalent to the local model $f(x_0+h) = f(x_0) + f'(x_0)h + o(|h|)$.

Secant slopes approach the tangent slope

The secant slope is

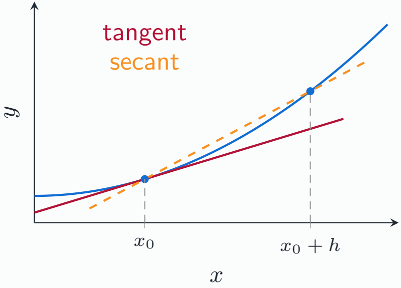
$$m_h = \frac{f(x_0+h) - f(x_0)}{h}.$$

As $h \to 0$, $m_h \to f'(x_0)$.

DEFINITION — Tangent line

$$y = f(x_0) + f'(x_0)(x - x_0).$$

It is the first-order affine approximation.



Differentiability is stronger than continuity

PROPOSITION — Differentiability implies continuity

If f is differentiable at x_0, then f is continuous at x_0.

Proof idea. $f(x_0 + h) - f(x_0) = h[f'(x_0) + o(1)] \to 0.$

EXAMPLE — The converse fails

For $f(x) = |x|$ at zero,

$$\lim_{h \to 0^-} \frac{|h|}{h} = -1, \qquad \lim_{h \to 0^+} \frac{|h|}{h} = 1.$$

The function is continuous, but the two candidate slopes disagree.

Differentiation rules reduce complex calculations

For differentiable functions on a common open interval,

$$(af + bg)' = af' + bg', \qquad (fg)' = f'g + fg'.$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \qquad \text{where } g \neq 0.$$

$f(x)$	$x^\alpha \ (x > 0)$	e^x	$\log x \ (x > 0)$
$f'(x)$	$\alpha x^{\alpha-1}$	e^x	$1/x$

EXAMPLE — Product rule

$$\left(\tfrac{2}{3}x^3 \log|x|\right)' = 2x^2 \log|x| + \tfrac{2}{3}x^2 \quad (x \neq 0).$$

The chain rule multiplies local sensitivities

PROPOSITION — Chain rule

If g is differentiable at x_0 and f at $g(x_0)$, with the composition defined nearby,

$$(f \circ g)'(x_0) = f'(g(x_0))g'(x_0).$$

Proof idea. Write $\Delta = g(x_0 + h) - g(x_0) = g'(x_0)h + o(|h|)$. Differentiability of f gives

$$f(g(x_0) + \Delta) - f(g(x_0)) = f'(g(x_0))\Delta + o(|\Delta|).$$

Since $\Delta = O(|h|)$, the last remainder is $o(|h|)$, proving the formula.

EXAMPLE — Two successive changes

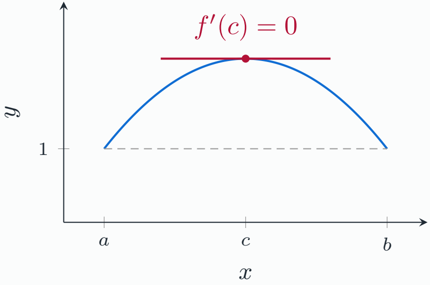
$\frac{\mathrm{d}}{\mathrm{d}x} e^{x^2} = e^{x^2} \cdot 2x.$

Rolle's theorem locates a horizontal tangent

THEOREM — Rolle

Let $a < b$. If f is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then some $c \in (a, b)$ satisfies $f'(c) = 0$.

Proof idea. Weierstrass gives extrema. Unless f is constant, one extremum is interior. The one-sided difference quotients there have opposite weak signs, forcing derivative zero.



The mean value theorem links local and finite change

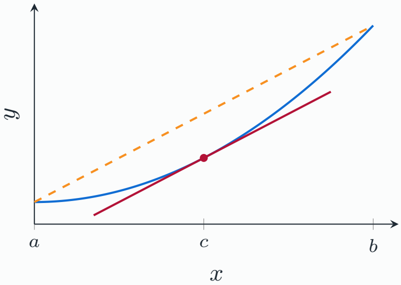
THEOREM — Mean value theorem

For continuous $f : [a, b] \to \mathbb{R}$, $a < b$, differentiable on (a, b),

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some $c \in (a, b)$.

Proof idea. Subtract the secant line, then apply Rolle's theorem.



$f(x) = 0.2x^2 + 0.4$ on $[0, 4]$: $c = 2$, slope 0.8.

Derivative bounds control output perturbations

PROPOSITION — A quantitative stability bound

If f is differentiable on an open interval I and $|f'| \le L$, then

$$|f(y) - f(x)| \le L|y - x| \qquad (x, y \in I).$$

Proof idea. Apply the mean value theorem on the segment between x and y.

EXAMPLE — Error propagation

If an estimate has error $|\widehat{\theta} - \theta| \le \delta$, then $|f(\widehat{\theta}) - f(\theta)| \le L\delta$ when the derivative bound holds along the segment.

In machine learning, analogous gradient bounds control sensitivity of predictions and losses.

The derivative sign determines monotonicity

PROPOSITION — On an open interval I

$f' \geq 0$ everywhere implies f is increasing.
$f' > 0$ everywhere implies f is strictly increasing.
$f' = 0$ everywhere implies f is constant.

Proof idea. For $x < y$, $f(y) - f(x) = f'(c)(y - x)$ for some $c \in (x, y)$.

Check the assumptions

Strict increase does not require a strictly positive derivative at every point: $f(x) = x^3$ is strictly increasing, but $f'(0) = 0$.

Reverse the signs for decreasing functions.

Taylor–Young adds higher-order local corrections

DEFINITION — Little-o remainder

$r(h) = o(|h|^n)$ means $r(h)/|h|^n \to 0$ as $h \to 0$, $h \neq 0$.

THEOREM — Taylor–Young

For $f \in C^n(I)$, I open, $x_0 \in I$ and $n \geq 1$,

$$f(x_0 + h) = \sum_{j=0}^{n} \frac{f^{(j)}(x_0)}{j!} h^j + o(|h|^n).$$

Proof idea. The case $n = 1$ is differentiability. Inductively, the remainder R satisfies $R(0) = 0$ and $R'(h) = o(|h|^{n-1})$. The MVT gives $R(h) = hR'(\xi_h) = o(|h|^n)$.

Standard expansions reveal the leading terms

As $h \to 0$,

$$e^h = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + o(|h|^3),$$

$$\sin h = h - \frac{h^3}{6} + o(|h|^3), \qquad \cos h = 1 - \frac{h^2}{2} + o(h^2),$$

$$\log(1+h) = h - \frac{h^2}{2} + o(h^2),$$

$$(1+h)^\alpha = 1 + \alpha h + \frac{\alpha(\alpha-1)}{2} h^2 + o(h^2).$$

Coefficients are derivatives evaluated at the expansion point. For real powers and logarithms, $1 + h > 0$ near zero.

Local polynomials resolve cancellations

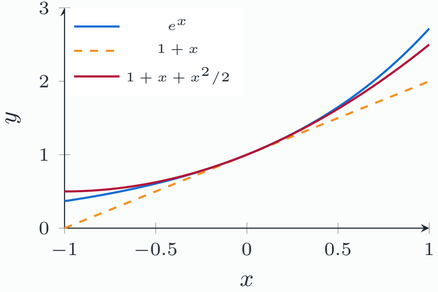
EXAMPLE — An indeterminate quotient

$$\frac{e^h - 1 - h}{h^2} = \frac{1}{2} + o(1) \to \frac{1}{2}.$$

The constant and linear terms cancel; the quadratic term decides the limit.

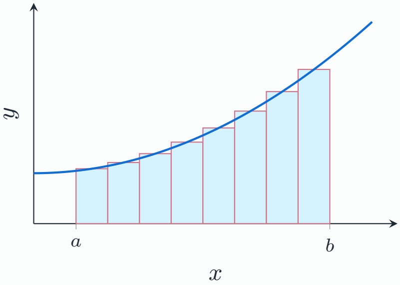
Check the assumptions

Taylor–Young is local. Its asymptotic remainder alone does not supply a numerical error bound at a prescribed finite distance.

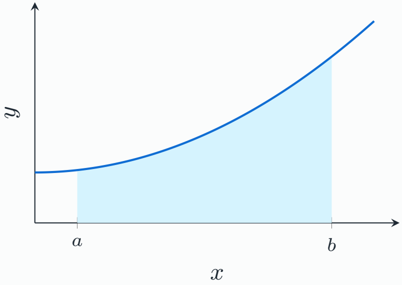


Course Outline

Equal-width rectangles approximate an area



$S_N = \sum_{i=1}^{N} f(\xi_i)\Delta x$



$\int_a^b f(x)\,\mathrm{d}x$

Here $f(x) = 0.8 + 0.15x^2$ on $[0.5, 3.5]$; each rectangle uses its midpoint height. As the mesh tends to zero, the Riemann sums converge to the integral.

The Riemann integral is independent of the sampling tags

DEFINITION — Tagged partitions

For $a = x_0 < \cdots < x_N = b$, choose $\xi_i \in [x_{i-1}, x_i]$. The mesh is $|P| = \max_i(x_i - x_{i-1})$ and

$$S(f, P) = \sum_{i=1}^{N} f(\xi_i)(x_i - x_{i-1}).$$

For continuous f, these sums have a common limit as $|P| \to 0$, independent of partitions and tags.

Proof idea. Uniform continuity bounds the difference between upper and lower sums by $(b - a)\omega_f(|P|) \to 0$.

Integration sums local contributions in a way that stabilizes under finer discretization.

An integral is signed accumulation

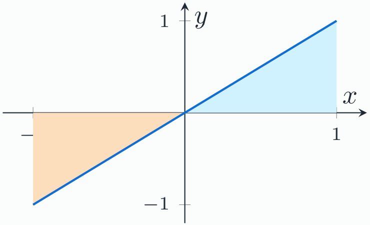
$$\int_a^b f = \text{area above axis} - \text{area below axis.}$$

Geometric area is $\int_a^b |f|$.

EXAMPLE — Cancellation

$\int_{-1}^{1} x \, \mathrm{d}x = 0$, although $\int_{-1}^{1} |x| \, \mathrm{d}x = 1$.

Net change and total accumulated magnitude are different.



Integrating a rate recovers a net change

Suppose $F \in C^1([a,b])$ and $f = F'$ is its instantaneous rate.

$$F(b) - F(a) = \sum_i \big(F(t_i) - F(t_{i-1})\big).$$

On each subinterval, the MVT gives $F(t_i) - F(t_{i-1}) = f(\xi_i)\Delta t_i$. Passing to the limit,

net change $$F(b) - F(a) = \int_a^b f(t)\,\mathrm{d}t.$$

EXAMPLE — A quantity growing at rate $1 + t/2$

If $F(0) = 2$, then $F(t) = 2 + t + t^2/4$ and $F(3) - F(0) = 21/4$.

Antiderivatives differ only by a constant

DEFINITION — Antiderivative on an open interval

F is an antiderivative of f on I if $F'(x) = f(x)$ for every $x \in I$.

PROPOSITION — Uniqueness up to a constant

If $F' = G' = f$ on I, then $F - G = C$ on I.

Proof idea. The derivative of $F - G$ is zero; the MVT makes $F - G$ constant.

EXAMPLE — An initial condition selects one primitive

$F' = 2x$ gives $F(x) = x^2 + C$. Requiring $F(0) = 3$ fixes $C = 3$.

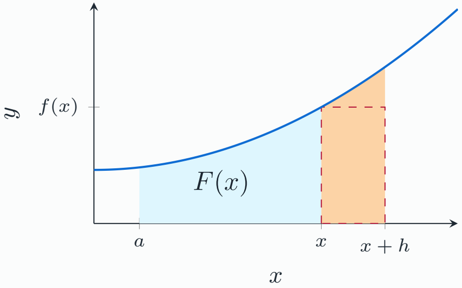
Differentiating accumulation recovers the integrand

THEOREM — Fundamental theorem, Part I

For continuous f on an interval, define $F(x) = \int_a^x f(t)\,\mathrm{d}t$. At every interior point, $F'(x) = f(x)$.

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,\mathrm{d}t.$$

The quotient is an average height.



Orange strip: added accumulation. Dashed rectangle: height $f(x)$.

Continuity is what controls the shrinking strip

For $h > 0$,

$$\left|\frac{F(x+h)-F(x)}{h} - f(x)\right| = \left|\frac{1}{h}\int_x^{x+h} (f(t)-f(x))\,\mathrm{d}t\right|$$
$$\leq \sup_{t\in[x,x+h]} |f(t)-f(x)| \longrightarrow 0.$$

The case $h < 0$ uses the interval $[x+h, x]$ and the orientation of the integral.

The strip area tends to zero. Its area divided by its width tends to $f(x)$, because continuity makes nearby heights uniformly close to $f(x)$.

A primitive evaluates a definite integral at its endpoints

THEOREM — Fundamental theorem, Part II

If f is continuous on $[a, b]$ and F is a primitive (continuous at the endpoints), then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Proof idea. The accumulation $G(x) = \int_a^x f$ satisfies $G' = f = F'$. Thus $F - G$ is constant, and $G(a) = 0$ gives the formula.

EXAMPLE — Polynomial integration

$$\int_0^2 (3x^2 + 1)\,\mathrm{d}x = [x^3 + x]_0^2 = 10.$$

Integration respects linearity, orientation and order

For continuous functions on the intervals involved,

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g,$$

$$\int_a^c f = \int_a^b f + \int_b^c f, \qquad \int_b^a f = -\int_a^b f.$$

PROPOSITION — Order and bounds, with $a < b$

If $m \leq f(x) \leq M$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a).$$

In particular, $\left|\int_a^b f\right| \leq \int_a^b |f|$.

Integration by parts reverses the product rule

PROPOSITION — For $u, v \in C^1([a,b])$

$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x.$$

Proof idea. Integrate $(uv)' = u'v + uv'$ and use the fundamental theorem.

EXAMPLE — A polynomial times an exponential

Choose $u = x$, $v' = e^x$, so $u' = 1$, $v = e^x$:

$$\int_0^1 xe^x\,\mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x\,\mathrm{d}x = e - (e-1) = 1.$$

Choose the factors so that differentiating one simplifies the remaining integral.

Course Outline

Optimization asks for the best attainable value

DEFINITION — Global minimum on D

$x^* \in D$ is a global minimizer if $f(x^*) \leq f(x)$ for every $x \in D$. The minimum value is $f(x^*)$; a minimizer is an input attaining it.

EXAMPLE — A least-squares objective

Given observations $y_1, \ldots, y_m$, choose θ to minimize

$$L(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (\theta - y_i)^2.$$

First ask whether an optimum exists. Then characterize candidates and determine whether they are local or global.

Interior differentiable extrema are stationary

PROPOSITION — Fermat's necessary condition

If f has a local extremum at an interior point c and is differentiable there, then $f'(c) = 0$.

Proof idea. At a local maximum, positive-h difference quotients are nonpositive and negative-h quotients are nonnegative. Their common limit is zero.

Check the assumptions

The converse fails: x^3 has derivative zero at 0 but no extremum there. Nondifferentiable points and domain endpoints must also be examined.

For a differentiable function, a stationary point is a point where its derivative vanishes.

On a compact interval, compare all candidate values

EXAMPLE — Minimize and maximize $f(x) = x^2 - 2x + 3$ on $[0, 4]$

Continuity and compactness guarantee attained extrema.

$$f'(x) = 2x - 2 = 0 \iff x = 1.$$

$$f(0) = 3, \qquad f(1) = 2, \qquad f(4) = 11.$$

Therefore $\min f = 2$ at 1 and $\max f = 11$ at 4.

Find interior stationary or nondifferentiable candidates, include the endpoints, and compare their values. If there are infinitely many candidates, analyze the candidate set.

A derivative sign change classifies local extrema

Suppose f is continuous at c and differentiable on either side.

Left of c	Right of c	Conclusion
$f' < 0$	$f' > 0$	Strict local minimum
$f' > 0$	$f' < 0$	Strict local maximum

EXAMPLE — The function $x \log x$ on $(0, +\infty)$

$f'(x) = \log x + 1$ changes from negative to positive at $x = e^{-1}$. Thus $f(e^{-1}) = -e^{-1}$ is the global minimum, since the sign pattern holds throughout the domain.

Convexity makes local minima global

DEFINITION — Convex function on an interval

$$f((1-t)x+ty) \le (1-t)f(x)+tf(y), \qquad t \in [0,1].$$

The graph lies below every chord. Concavity reverses the inequality.

For twice differentiable f on an open interval, convexity is equivalent to $f'' \ge 0$.

PROPOSITION — Global significance

Every local minimum of a convex function is global. Strict convexity gives at most one minimizer.

Proof idea. If another point had a smaller value, convexity would give smaller values arbitrarily close to the supposed local minimizer.

The second derivative detects a nondegenerate extremum

For $f \in C^2$ near c with $f'(c) = 0$,

$$f(c+h) - f(c) = \tfrac{1}{2} f''(c)h^2 + o(h^2).$$

PROPOSITION — Second derivative test

$f''(c) > 0$ gives a strict local minimum; $f''(c) < 0$ gives a strict local maximum. If $f''(c) = 0$, the test is inconclusive.

EXAMPLE — The same zero second derivative, different outcomes

At 0, x^4 has a minimum, $-x^4$ has a maximum, and x^3 has neither.

The nonzero quadratic term dominates the remainder for sufficiently small h.

A revenue model combines local calculus and global analysis

Suppose unit price is $p(x) = e^{-2x}$ for quantity $x \geq 0$. Revenue is $R(x) = xe^{-2x}$.

$$R'(x) = (1 - 2x)e^{-2x}.$$

It is positive for $0 \leq x < 1/2$ and negative for $x > 1/2$.

EXAMPLE — The optimal decision

The unique global maximizer is $x^* = 1/2$, with price $p(x^*) = e^{-1}$ and revenue $R(x^*) = 1/(2e)$.

The sign pattern on the entire feasible domain proves global optimality; a negative second derivative at one point would only prove local optimality.

Course Outline

Several inputs produce surfaces and level sets

DEFINITION — Scalar function of several variables

$f : D \subseteq \mathbb{R}^d \to \mathbb{R}$, with inputs represented as column vectors.

$$\|x - y\| = \sqrt{\sum_{j=1}^{d} (x_j - y_j)^2}.$$

In two variables, the graph is $\{(x, y, f(x, y)) : (x, y) \in D\} \subseteq \mathbb{R}^3$.

EXAMPLE — A domain restriction

$f(x, y) = \sqrt{y - x}$ has domain $D = \{(x, y) : y \geq x\}$ and image $[0, +\infty)$.

A level set $L_k = \{x \in D : f(x) = k\}$ lies in input space, not in graph space.

Partial derivatives examine coordinate directions

DEFINITION — At an interior point of the domain

$$\frac{\partial f}{\partial x}(x,y) = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}, \qquad \frac{\partial f}{\partial y}(x,y) = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}.$$

EXAMPLE — A quadratic function

For $f(x,y) = x^2 + xy + 2y^2$,

$$\frac{\partial f}{\partial x} = 2x + y, \qquad \frac{\partial f}{\partial y} = x + 4y, \qquad \nabla f = \begin{bmatrix} 2x+y \\ x+4y \end{bmatrix}.$$

Continuity requires $f(x+h) \to f(x)$ along every vector approach $h \to 0$, not only along the axes.

Differentiability requires one model for all directions

DEFINITION — Differentiability at p in an open domain

$$f(p+h) = f(p) + Df(p)[h] + o(\|h\|), \qquad Df(p)[h] = \nabla f(p)^\top h.$$

Continuous first partial derivatives near p are sufficient.

EXAMPLE — Partial derivatives alone do not suffice

Set $f(0,0) = 0$ and $f(x,y) = xy/\sqrt{x^2+y^2}$ elsewhere. It is continuous at zero and both partial derivatives there vanish, but

$$\frac{f(t,t)}{\|(t,t)\|} = \frac{1}{2} \not\to 0.$$

When f is differentiable, the tangent plane is $z = f(p) + \nabla f(p)^\top (x-p)$.

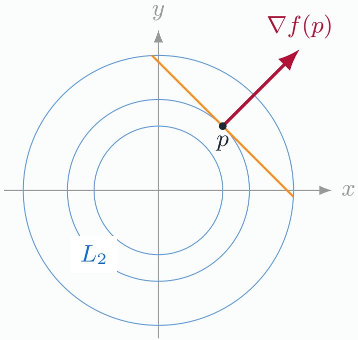
The gradient is normal to regular level sets

Let $f \in C^1(U)$, U open, and let $\gamma(t)$ lie in L_k with $\gamma(0) = p$.

PROPOSITION — Orthogonality

$$\nabla f(p)^\top \gamma'(0) = 0.$$

Proof idea. Differentiate the constant identity $f(\gamma(t)) = k$ using the chain rule. At a regular point, $\nabla f(p) \neq 0$, this vector specifies a normal direction.



$f(x, y) = x^2 + y^2$, $p = (1, 1)$.

The gradient points towards steepest first-order increase

For differentiable f and a unit vector v,

$$D_v f(p) = \nabla f(p)^\top v \le \|\nabla f(p)\|.$$

Cauchy–Schwarz gives equality at $v = \nabla f(p)/\|\nabla f(p)\|$ when the gradient is nonzero.

PROPOSITION — A local descent direction

$$f(p - t\nabla f(p)) = f(p) - t\|\nabla f(p)\|^2 + o(t) < f(p)$$

for sufficiently small $t > 0$ when $\nabla f(p) \neq 0$.

Check the assumptions

This is a local statement in the Euclidean norm. An arbitrary large step may increase the objective.

Jacobians compose local linear maps

For differentiable $g : \mathbb{R}^q \to \mathbb{R}^p$, the Jacobian is

$$J_g(x) = \left[\frac{\partial g_i}{\partial x_j}(x)\right]_{i,j} \in \mathbb{R}^{p\times q}.$$

THEOREM — Chain rule for scalar $f : \mathbb{R}^p \to \mathbb{R}$

$$\nabla(f \circ g)(x) = J_g(x)^\top \nabla f(g(x)).$$

For a curve $g : \mathbb{R} \to \mathbb{R}^p$, $(f \circ g)'(t) = \nabla f(g(t))^\top g'(t)$.

Proof idea. Substitute $g(x+h) = g(x) + J_g(x)h + o(\|h\|)$ into the linear approximation of f.

The multiplication of Jacobians is the mathematical basis of the chain of derivatives used in backpropagation.

Multivariable Taylor includes coordinate interactions

THEOREM — For $f \in C^n(U)$, U open, $x_0 \in U$

$$f(x_0+h) = \sum_{k=0}^{n} \frac{1}{k!} D^k f(x_0)[h,\ldots,h] + o(\|h\|^n).$$

The $j=0$ term is $f(x_0)$; $D^j f(x_0)$ is a symmetric j-linear map.

At order two, with Hessian $H_f = [\partial_i \partial_j f]$,

$$f(x_0+h) = f(x_0) + \nabla f(x_0)^\top h + \tfrac{1}{2} h^\top H_f(x_0) h + o(\|h\|^2).$$

EXAMPLE — Expansion at the origin

$e^{u+2v} = 1 + u + 2v + \frac{1}{2}u^2 + 2uv + 2v^2 + o(\|(u,v)\|^2)$. The mixed term $2uv$ records an interaction between inputs.

Global optimization: existence of extrema

DEFINITION — Global extrema

Let $D \subseteq \mathbb{R}^2$ be nonempty and let $f : D \to \mathbb{R}$. A point $p_{\min} \in D$ is a **global minimizer** if

$$f(p_{\min}) \le f(p) \qquad \text{for every } p \in D.$$

A **global maximizer** is defined by reversing the inequality.

Distinguish the **points** where extrema are attained from the corresponding **values**:

$$m = f(p_{\min}) = \min_D f, \qquad M = f(p_{\max}) = \max_D f.$$

THEOREM — Weierstrass

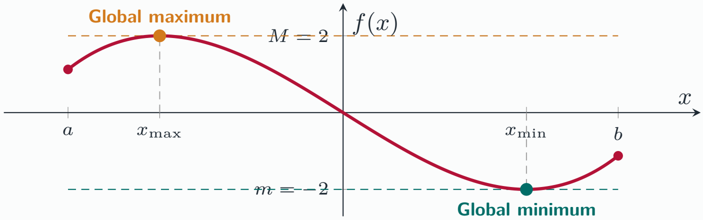
If D is **closed and bounded** and f is continuous on D, then f attains both a global minimum and a global maximum on D.

Existence is the starting point

The theorem guarantees that optimal points exist. It does not locate them or imply that they are

From existence to computation

A one-dimensional illustration: $f(x) = x^3 - 3x$ on the closed interval $D = [-\frac{3}{2}, \frac{3}{2}]$.



Locate candidates, then compare values

If f is differentiable in the interior, find the stationary points and examine the boundary separately. Here,

$$f'(x) = 3(x^2 - 1) = 0 \iff x = \pm 1, \qquad f(a) = \frac{9}{8}, \quad f(b) = -\frac{9}{8}.$$

Comparing these four values gives $\min_D f = f(1) = -2$ and $\max_D f = f(-1) = 2$.

Local extrema and stationary points

Assume that f is differentiable on an open set $U \subseteq \mathbb{R}^2$, and let $p \in U$.

DEFINITION — Local behaviour

The point p is a **local minimizer** if, for some $r > 0$,

$$B(p,r) \subseteq U, \qquad f(p) \leq f(q) \quad \text{for every } q \in B(p,r).$$

Reverse the inequality for a **local maximizer**. The extremum is **strict** if the inequality is strict whenever $q \neq p$.

PROPOSITION — Fermat: a necessary condition

Every local extremum in U is a **stationary point**:

$$\nabla f(p) = 0 \quad \iff \quad \frac{\partial f}{\partial x}(p) = \frac{\partial f}{\partial y}(p) = 0.$$

Why must the gradient vanish?

At an interior local extremum, restricting f to any line through p gives a differentiable function with a

A stationary point need not be an extremum

DEFINITION — Saddle point

A **saddle point** is a stationary point p such that every neighbourhood of p contains points q_- and q_+ with

$$f(q_-) < f(p) < f(q_+).$$

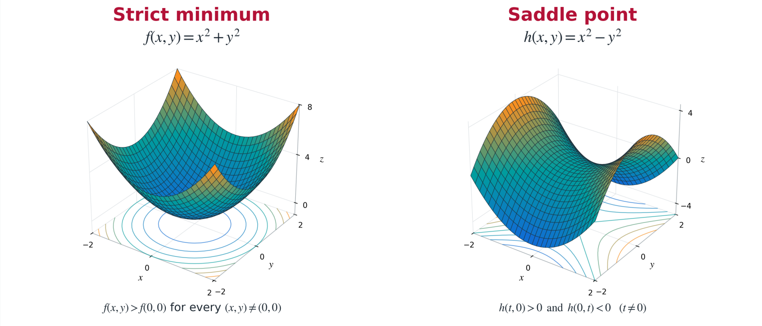


Figure: Both gradients vanish at the origin, but only the first point is a local minimum.

The Hessian describes curvature near a stationary point I

Assume $f \in C^2(U)$, where $U \subseteq \mathbb{R}^2$ is open, and let $p^* \in U$ satisfy $\nabla f(p^*) = 0$.

DEFINITION — Hessian matrix

The Hessian is the symmetric matrix

$$H = \nabla^2 f(p^*) = \begin{pmatrix} f_{xx}(p^*) & f_{xy}(p^*) \\ f_{xy}(p^*) & f_{yy}(p^*) \end{pmatrix}.$$

Its eigenvalues λ_1, λ_2 are real.

Why curvature determines the local behaviour

Taylor's formula at the stationary point gives

$$f(p^* + h) - f(p^*) = \frac{1}{2} h^\top H h + o(\|h\|^2).$$

The quadratic term controls the sign when H is definite; opposite curvature signs produce a saddle.

The Hessian describes curvature near a stationary point II

PROPOSITION — Second-order classification

Eigenvalues of H	Conclusion at p^*
$\lambda_1 > 0,\ \lambda_2 > 0$	Strict local minimum
$\lambda_1 < 0,\ \lambda_2 < 0$	Strict local maximum
$\lambda_1\lambda_2 < 0$	Saddle point
$\lambda_1\lambda_2 = 0$	Test inconclusive

A practical test without computing eigenvalues I

For $f \in C^2(U)$ and an interior stationary point p^*, write

$$H = \nabla^2 f(p^*) = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad \Delta = \det H = ac - b^2.$$

PROPOSITION — The two-variable Hessian test

Condition	**Conclusion at p^***
$\Delta > 0$ and $a > 0$	Strict local minimum
$\Delta > 0$ and $a < 0$	Strict local maximum
$\Delta < 0$	Saddle point
$\Delta = 0$	Test inconclusive

The identities

$$\lambda_1 \lambda_2 = \Delta, \qquad \lambda_1 + \lambda_2 = \operatorname{tr} H = a + c$$

explain the test: if $\Delta > 0$, the eigenvalues have the same sign, and a, c share that sign.

A practical test without computing eigenvalues II

EXAMPLE — A zero Hessian does not determine the outcome

At the origin, all three functions below have $\nabla f = 0$ and $H = 0$:

$x^4 + y^4$	$-x^4 - y^4$	$x^4 - y^4$
strict minimum	strict maximum	saddle

Higher-order terms or direct comparisons are then needed.

Summary: The Hessian classifies stationary points

Assume $f \in C^2(U)$ and $p \in U$ with $\nabla f(p) = 0$. Then $f(p+h) - f(p) = \frac{1}{2} h^\top H_f(p) h + o(\|h\|^2)$.

Hessian	Eigenvalues	Conclusion
Positive definite	All positive	Strict local minimum
Negative definite	All negative	Strict local maximum
Indefinite	Positive and negative	Saddle point
Singular semidefinite	Zero eigenvalue(s)	Test inconclusive

Proof idea. If $H \succ 0$, then $h^\top H h \geq \lambda_{\min} \|h\|^2$ with $\lambda_{\min} > 0$. This dominates the remainder. Opposite eigenvalue signs give opposite variations along eigenvectors.

Least squares connects analysis to linear algebra

For $A \in \mathbb{R}^{m\times d}$ and $b \in \mathbb{R}^m$, consider $L(x) = \frac{1}{2}\|Ax - b\|^2$.

$$\nabla L(x) = A^\top (Ax - b), \qquad H_L(x) = A^\top A.$$

$$L(x+h) = L(x) + \nabla L(x)^\top h + \tfrac{1}{2}\|Ah\|^2.$$

PROPOSITION — Optimality and uniqueness

The normal equations $A^\top Ax = A^\top b$ characterize global minimizers. Since $A^\top A \succeq 0$, the loss is convex. Full column rank makes it strictly convex and the minimizer unique.

Without full rank, minimizers still exist, but displacements in $\ker A$ leave the loss unchanged.

A two-product revenue model has a unique optimum

Unit prices are $p(x, y) = 100 - 3x - y$ and $q(x, y) = 180 - x - 4y$.

$$R(x, y) = xp + yq = 100x + 180y - 3x^2 - 2xy - 4y^2.$$

$$\nabla R(x^\star, y\star) = \begin{bmatrix} 100 - 6x^\star - 2y^\star \\ 180 - 2x^\star - 8y^\star \end{bmatrix} = 0 \iff (x^\star, y^\star) = (10, 20).$$

$$H_R = \begin{bmatrix} -6 & -2 \\ -2 & -8 \end{bmatrix}, \qquad \det H_R = 44 > 0.$$

The Hessian is negative definite everywhere, so R is strictly concave.

The unique global maximizer is $(10, 20)$, with prices $(50, 90)$ and revenue 2300. It is feasible under $x, y, p, q \geq 0$.

Analysis connects existence, sensitivity and computation

Mathematical tool	What it establishes
Completeness and monotonicity	Convergence of bounded monotone approximations
Continuity and compactness	Existence of attained extrema
Contraction	A unique fixed point and a convergent iteration
Derivatives and Taylor	Local sensitivity and controlled asymptotic models
Integration	Accumulation of local contributions
Convexity and Hessians	Global structure and local classification of optima

Check the domain and hypotheses. Distinguish existence, uniqueness and algorithmic convergence; distinguish local information from global guarantees.